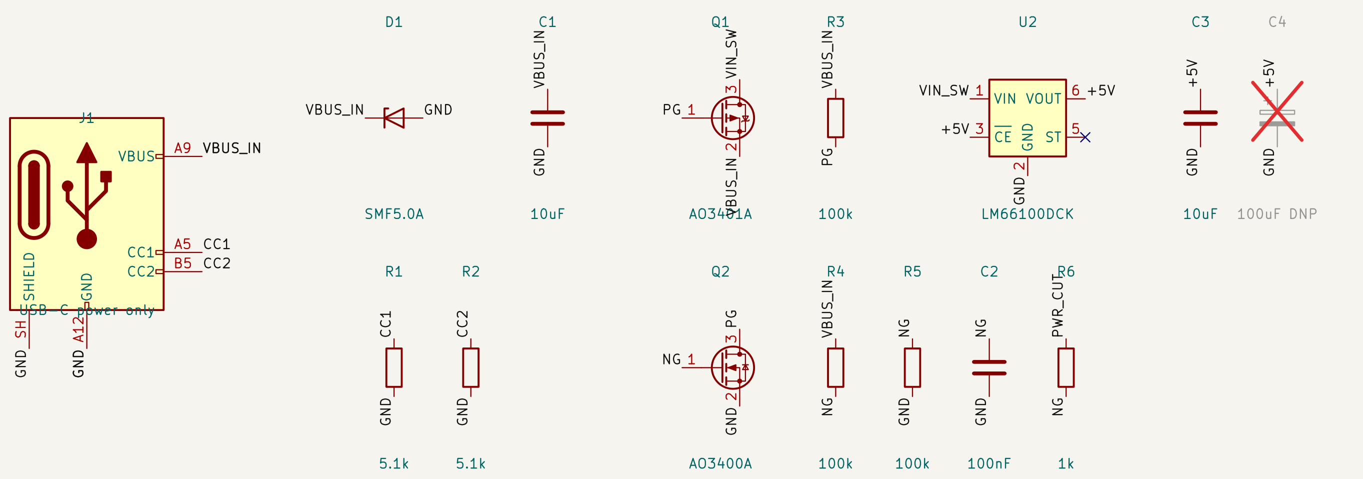
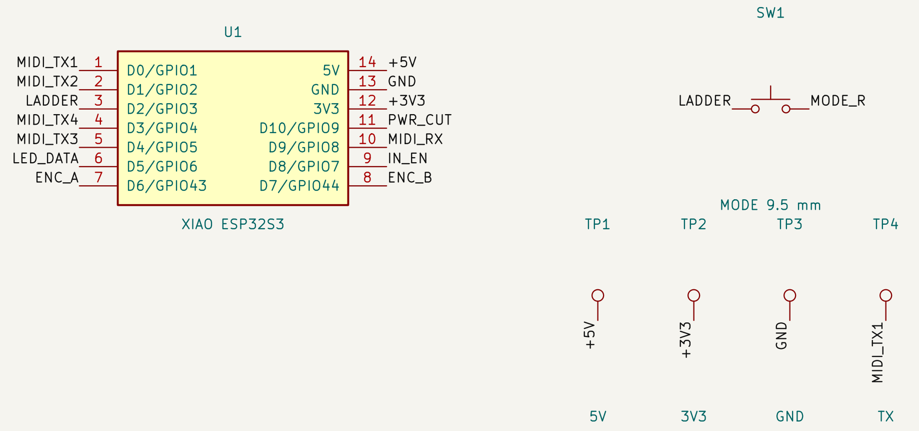
POWER IN: J1 power only (D+/D− not fitted). Q1 cut (on by default), U2 ideal diode onto the XIAO's 5V

U2 ~CE tied to VOUT: the datasheet's ideal–diode mode (blocks current back into J1). The XIAO's 5V pin IS its USB–C VBUS (no part between, Seeed v1.2 schematic): a computer on the XIAO port and a charger on J1 share +5V. Firmware pulls PWR_CUT low once a computer has enumerated it (device mode): Q2 off, Q1 off, J1 cut. At reset PWR_CUT floats: R4/R5 hold Q2 on, so a charger alone always starts it. C4 (100 uF) not fitted: host–mode hot–plug reserve, against USB's 10 uF limit for a device.


J1
VBUS
A9 VBUS_IN
A5 CC1
B5 CC2
SHIELD
GND
USB-C power only
D1
VBUS_IN
GND
SMF5.0A
C1
10uF
Q1
PG
VIN_SW
AO3401A
R3
100k
U2
VIN_SW
+5V
VIN VOUT
CE GND ST
LM66100DCK
C3
10uF
C4
100uF DNP
R1
CC1
5.1k
R2
CC2
5.1k
Q2
NG
AO3400A
R4
100k
R5
100k
C2
100nF
R6
PWR_CUT
1k

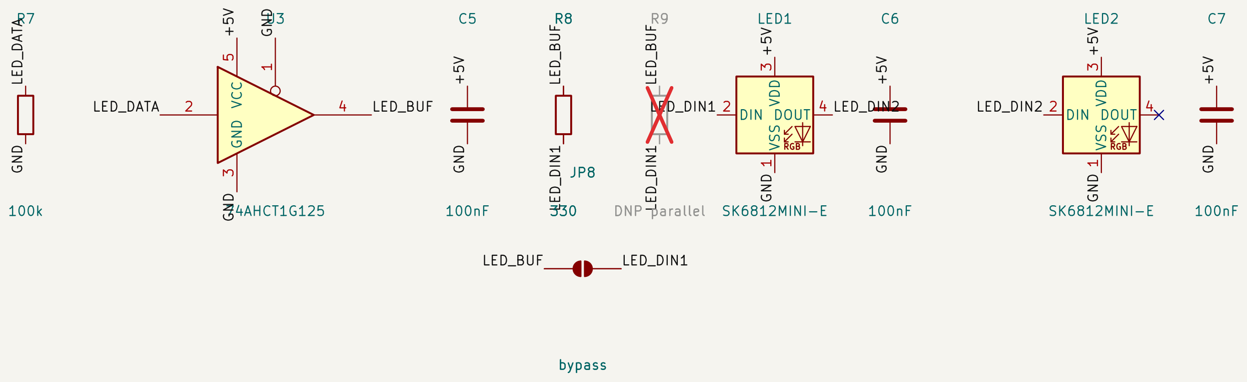
XIAO ESP32S3 (Seeed footprint, 2 x 7 headers or castellations)

D6 = GPIO43 has 499 ohm on the module (U0TXD). D2 = GPIO3 strapping, ignored by default eFuses. ENC_A = D6 = GPIO43, ENC_B = D7 = GPIO44 (U0TXD / U0RXD), IN_EN = D8 = GPIO7, MIDI_RX = D9 = GPIO8.
MIDI_TX1..4 = D0 D1 D4 D3 = GPIO1 2 5 4: UART TX routed to all four by the GPIO matrix (firmware).
LADDER = D2 = GPIO3 (ADC1): MODE / PLAY / PUSH on one resistor ladder. LED_DATA = D5 = GPIO6. PWR_CUT = D10 = GPIO9, low = J1 cut.


U1
MIDI_TX1 1 D0/GPIO1
MIDI_TX2 2 D1/GPIO2
LADDER 3 D2/GPIO3
MIDI_TX4 4 D3/GPIO4
MIDI_TX3 5 D4/GPIO5
LED_DATA 6 D5/GPIO6
ENC_A 7 D6/GPIO43
5V 14 +5V
GND 13 GND
3V3 12 +3V3
D10/GPIO9 11 PWR_CUT
D9/GPIO8 10 MIDI_RX
D8/GPIO7 9 IN_EN
D7/GPIO44 8 ENC_B
XIAO ESP32S3
SW1
LADDER
MODE_R
MODE 9.5 mm
TP1
+5V
5V
TP2
+3V3
3V3
TP3
GND
GND
TP4
MIDI_TX1
TX

LEDS: SK6812MINI–E x 2 on +5V; U3 lifts the 3.3 V data to 5 V (VIH = 0.65 x VDD = 3.25 V at 5 V)

GND
VBUS_IN
VIN_SW


R7
LED_DATA
100k
U3
LED_DATA 2
4 LED_BUF
74AHCT1G125
C5
100nF
R8
LED_BUF
330
JP8
R9
DNP parallel
LED1
LED_DIN1
DIN DOUT
LED_DIN2
SK6812MINI–E
C6
100nF
LED2
LED_DIN2
SK6812MINI–E
C7
100nF
LED_BUF
LED_DIN1
bypass

On the board each pair is ONE footprint (J6..J9; long board jank:MIDI_OUT_SDS–50J_or_SJ1–3523N_AB: the jack's pads T, R, S carry J2..J5's nets; square board: numbered 5, 4, 2 inside the DIN's). J2..J5 are BOM lines only (not on the board).

MIDI OUT 1–4 (CA–033 at 3.3 V): per output fit ONE of the two, the TRS (type A) OR the DIN–5

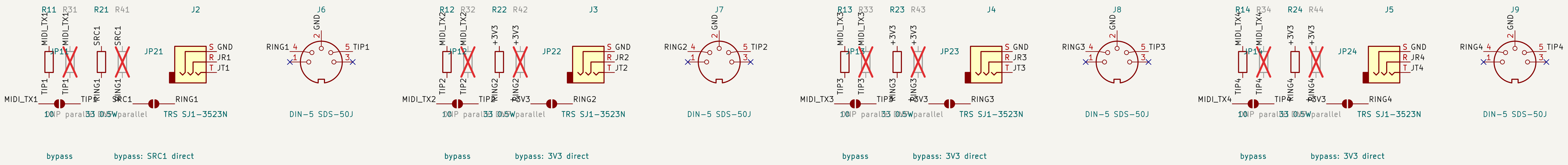
TRS type A: tip = data (10 ohm to the TX pad), ring = source (33 ohm 0.5 W to 3V3), sleeve = GND. DIN–5: pin 5 = tip net, pin 4 = ring net, pin 2 = GND, pins 1 and 3 open.

OUT1: J2 TRS or J6 DIN

OUT2: J3 TRS or J7 DIN

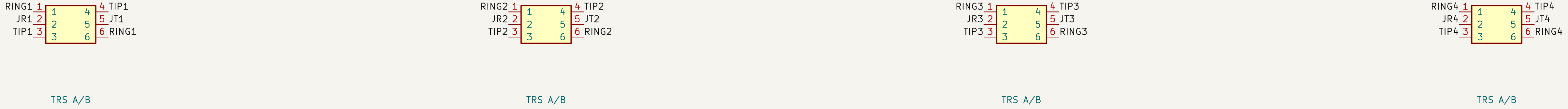
OUT3: J4 TRS or J8 DIN

OUT4: J5 TRS or J9 DIN


R11 R31 R21 R41
JP11 JP21
MIDI_TX1
TIP1
SRC1
RING1
10
DNP parallel
33 0.5W
bypass
bypass: SRC1 direct
J2
S GND
R JR1
T JT1
TRS SJ1–3523N
J6
RING1
TIP1
DIN–5 SDS–50J
R12 R32 R22 R42
JP12 JP22
MIDI_TX2
TIP2
+3V3
RING2
bypass: 3V3 direct
J3
TRS SJ1–3523N
J7
DIN–5 SDS–50J
R13 R33 R23 R43
JP13 JP23
MIDI_TX3
TIP3
RING3
J4
J8
R14 R34 R24 R44
JP14 JP24
MIDI_TX4
TIP4
RING4
J5
J9

SW2
SW3
SW4
SW5

Each MIDI resistor (and R8): axial, hand–swappable; JP across it = bypass (0 ohm, open by default); R3k / R4k / R9 in parallel, not fitted (lower the value). JP2k bridged = RING on 3V3 directly (JP21: on SRC1): a mono plug then shorts the rail.

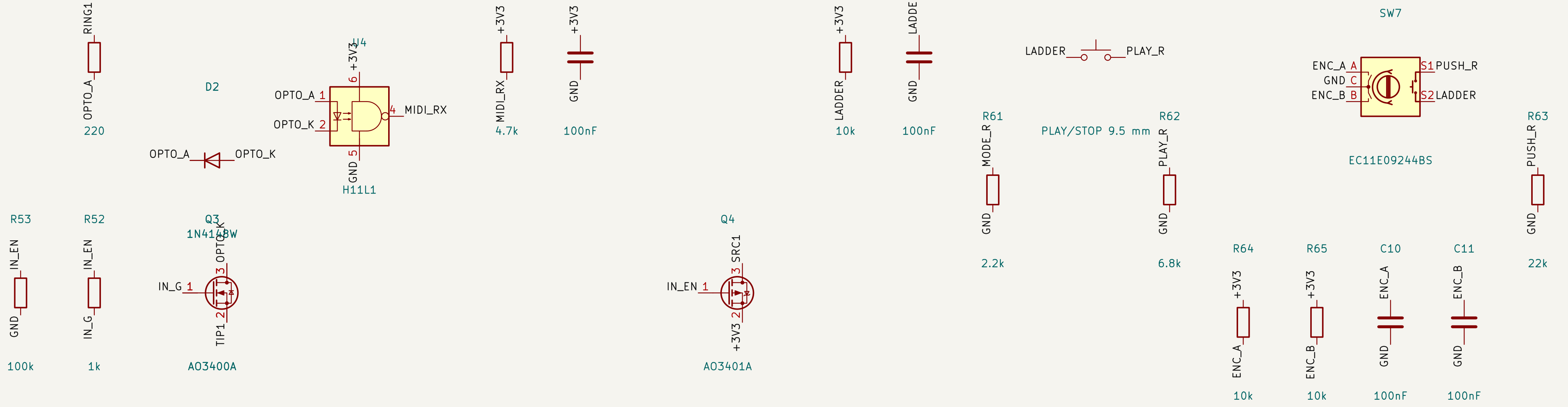

RING1 1
JR1 2
TIP1 3
4 TIP1
5 JT1
6 RING1
TRS A/B
RING2
JR2
TIP2
JT2
RING3
JR3
TIP3
JT3
RING4
JR4
TIP4
JT4

SW2..SW5: TRS type A / B per output (face side): they swap the jack's T and R between TIPk (data, sink) and RINGk (source) after the resistors; the DIN keeps pin 5 = TIPk, pin 4 = RINGk. 1–2 / 4–5 closed = type A.

OUT1 AS MIDI IN (software): TX1 to input (Hi–Z), IN_EN high (Q4 off: no sour[illegible]

CONTROLS (face side): [illegible] ladder on GPIO3: R60 10k to 3V3; MODE 2.2k, PLAY 6.8k, PUSH 22k to GND

OUT mode: IN_EN low (R53 at reset): Q4 on (3V3 to the 33 ohm), Q3 off, the opto's loop open (its body diode blocks). [illegible] to GND each.


R50
RING1
OPTO_A
220
D2
OPTO_A
OPTO_K
1N4148W
U4
OPTO_A
OPTO_K
MIDI_RX
H11L1
R51
+3V3
MIDI_RX
4.7k
C8
100nF
R53
IN_EN
100k
R52
IN_G
1k
Q3
AO3400A
Q4
IN_EN
SRC1
AO3401A
R60
LADDER
10k
C9
100nF
SW6
LADDER
PLAY_R
PLAY/STOP 9.5 mm
R61
MODE_R
2.2k
R62
PLAY_R
6.8k
SW7
ENC_A
GND
ENC_B
PUSH_R
LADDER
EC11E09244BS
R63
PUSH_R
22k
R64
ENC_A
10k
R65
ENC_B
10k
C10
100nF
C11
100nF

generated by make_schematic.py; do not edit by hand
hardware/pcb/README.md: the power path, the pin table and every unverified item

Sheet: /
File: jank–pcb.kicad_sch

Title: jank: XIAO ESP32S3 MIDI clock stick

Size: A2
Date: 2026–11–08
Rev: 0.1 schematic for review
KiCad E.D.A. 10.0.8
Id: 1/1